PACLAND · Drug discovery

The verification discipline applied to computational drug discovery — a plain-English brief

The headlines — and what to do about each

- A novel molecule is extrapolation, not verification (0 of 5 endpoints measured) — assay it before you bet, don't trust the in-silico score.
- Only 4 of 10 benchmark endpoints are validated — spend on assay data, not on more ML.
- The field's own census: 52% of 'verified' hits fail REOS, 18% are PAINS — rule out before you rank.
- Trust by verdict, not by confidence score — pLDDT is not calibrated.

1 · What we are measuring

Drug discovery is a search over chemical space (~10^{60} molecules) with scarce ground truth: an assay exists only for the molecules you have already tested. The discipline asks how much of a screening decision is verified, versus predicted.

2 · The oracle and the Rule of Three

- Oracle: chemical validity — a molecule must be valency/charge/tautomer-consistent (RDKit sanitisation). A 'molecule' with a pentavalent carbon is not a molecule; validity is an always-true answer.
- Rule of Three: three independent checks a candidate must all pass — validity, drug-likeness (Lipinski / PAINS / REOS), and synthesizability (RAScore).

3 · The verification deficit

- 4 of 10 benchmark endpoints are measured + computable + validated — a certificate radius of 68%.
- 0 of 5 endpoints for a novel molecule are validated — the hit you would synthesize is extrapolation, not verification.

4 · The constraint guard-suite

- 4 guards, each mutation-tested: net charge, valence bounds, Lipinski (rule of five), synthesizability.

The field has run these 'forbidden-state' filters for 25 years — PAINS, REOS, Lipinski, RDKit sanitization — as boolean heuristics. The guard-suite wraps them as a pass/fail test a model must pass, with a seeded fault proving each guard fires.

5 · Why it matters

The field's own 2013 census (Zhu et al.) found 52% of reported virtual-screening hits fail REOS and 18% are PAINS artifacts, and AlphaFold's confidence score (pLDDT) is not calibrated. 'Verified' hits are, in large part, artifacts — and the discipline makes that a machine-checked verdict.

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